For a function between sets the kernel relation is effective

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'prop:kernels_are_effective' (pinned-original-source.tex lines 385-389, original Proposition 10.1.9); the proof environment starts at line 391, one line after the blank line that closes the statement, so the pairing is by adjacency.

Editorial note. The retained counter values (chapter 10, section 1, theorem offset 8) make the counted proposition print as Proposition 10.1.9, the number the first-edition label-number dump gives for 'prop:kernels_are_effective'.

Editorial note. The only label of the retained range that is not defined inside it occurs on source line 394, which is commented out by the author; it is declared with 'non_load_bearing' true and with the author's own number, and the compiled card contains no reference to it.

Editorial note. The retained preamble is the mechanical standalone adapter followed by the book's own macro library, reproduced verbatim from the same pinned source tree.

Editorial scope. One original result of the pinned first-edition chapter, counted once: effectivity of the kernel relation of a function between sets (author label prop:kernels_are_effective).

Allowed standard prerequisites (editorial).

- Set-level quotient and effectivity. A relation R:A->A->prop is effective when the kernel pair of the quotient map is exactly the relation with respect to the quotient identification.. It is the property the statement claims for the kernel relation.
- Image as a set-quotient. im(f) is the set of pairs (b, |a,_|) with f(a)=b, i.e. the image of f as a subset of B.. The proof constructs the comparison map c_f:A->im(f) and studies its kernel pair.
- Kernel of a function. ker(f,x,y) is the proposition f(x)=f(y).. It is the relation that the theorem proves effective.

Source reference. Original reference 10.1.5: sources/hott-setmath/pinned-original-source.tex, source lines 394–394. Declared as a non-load-bearing navigation pointer only: the label occurs in a commented-out author line (line 394 begins with a percent sign), so the compiled card contains no reference to it. The number 10.1.5 is the author's own, taken from the first-edition label-number dump; it is the second label of the theorem stated at setmath.tex line 267.

Theorem 10.1.9. *For any function $f : A \to B$ between any two sets, the relation* $\ker(f) : A \to A \to \mathsf{Prop}$ *given by* $\ker(f, x, y) :\equiv (f(x) = f(y))$ *is effective.*

Proof. We will use that $\mathsf{im}(f)$ is the coequalizer of $\mathsf{pr}_1, \mathsf{pr}_2 : (\sum_{(x,y:A)} f(x) = f(y)) \to A$. Note that the kernel pair of the function

$$c_f :\equiv \lambda a.\, \Big(f(a), \big\|(a, \mathsf{refl}_{f(a)})\big\|\Big) : A \to \mathsf{im}(f)$$

consists of the two projections

$$\mathsf{pr}_1, \mathsf{pr}_2 : \Big(\sum_{x,y:A} c_f(x) = c_f(y)\Big) \to A.$$

For any $x, y : A$, we have equivalences

$$\begin{aligned}(c_f(x) = c_f(y)) &\simeq \Big(\sum_{p:f(x)=f(y)} p_*\big(\big\|(x, \mathsf{refl}_{f(x)})\big\|\big) = \big\|(y, \mathsf{refl}_{f(x)})\big\|\Big) \\ &\simeq (f(x) = f(y)),\end{aligned}$$

where the last equivalence holds because $\|\mathsf{fib}_f(b)\|$ is a mere proposition for any $b : B$. Therefore, we get that

$$\Big(\sum_{x,y:A} c_f(x) = c_f(y)\Big) \simeq \Big(\sum_{x,y:A} f(x) = f(y)\Big)$$

and hence we may conclude that $\ker f$ is an effective relation for any function f. □